


# Fundamentals of Cyber Security and Digital Forensics

## Module: I- Introduction of Cybersecurity

Sneha Mane

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# Teaching Scheme

| D Y PATIL DEEMED TO BE UNIVERSITY — RAMRAO ADIK — INSTITUTE OF TECHNOLOGY NAVI MUMBAI | | B.TECH IN INFORMATION TECHNOLOGY | | | | | | | | SECOND YEAR SEM- III | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Course Code** | **Course Name** | **Teaching Scheme (Contact Hours 39)** | | | | | | | | **Credits Assigned** | | |
| | | Theory | | | | | Practical | | | Theory | | Total |
| 2413CYM3T1 | Fundamentals of Cyber Security and Digital Forensics | 03 | | | | | -- | | | 03 | | **03** |
| | | **Theory Evaluation Scheme (Marks)** | | | | | | | | | | |
| | | Internal Assessment **(50)** | | | | | | | | End Sem. Exam | End Sem. Exam Hrs. | Total |
| | | Best 2 **(20)** | | | **(30)** | | | | | | | |
| | | CA1 | CA2 | CA3 | Assignment / Tutorial | | Quiz / Seminar | Open Book Test / Surprise Test / Capstone Project | | | | |
| | | 10 | 10 | 10 | 10 | | 10 | 10 | | **100** | 03 | **150** |



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## Course Pre-requisite:

- Computer Network, Operating System

## Course Description:

- The course is designed to turn the student into security expert and to prepare them for working as vulnerability researcher. Participants will learn how to find critical vulnerabilities on different platforms and exploit them, and will be given practical skills in core subjects of cyber security. The course includes in-depth knowledge of detection of cyber threats, its safeguards. Also it covers investigation process of cyber crime and understanding of cyber laws.



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## Course Objectives:

1. To understand the fundamentals of cyber security
2. To understand the background of cryptography
3. To analyze various mechanism of Cyber Security Safeguards
4. To understand the working of digital forensics tools
5. To discuss the knowledge of IT ACT and Cyber forensics laws



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## Course Outcome:

On successful completion of course, learner will be able to:

1. To understand breaching process of cyber threats.
2. To apply the knowledge of different cryptographic algorithm
3. To identify different Cyber attacks and apply Cyber Security mechanism.
4. To apply the principles of effective digital forensics investigation techniques.
5. To demonstrate digital forensics artifacts.
6. To Interpret and appropriately apply the laws and procedures associated with identifying, acquiring, examining and presenting digital evidence.



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## Syllabus :

| Module No. | Detailed Content | Hrs (39) | CO |
|---|---|---|---|
| 1 | Overview of Cyber Security, Internet Governance – Challenges and Constraints, Cyber Threats, Vulnerabilities and its types Need for a Comprehensive Cyber Security Policy, Need for a Nodal Authority, Need for an International convention on Cyberspace. | 6 | CO1 |
| 2 | Introduction to Cryptography, stream and block cipher, Symmetric key (DES and AES) and Asymmetric key Cryptography(RSA), Key exchange protocol(DH), Hash algorithm(MD5 and SHA), Message Authentication code(MAC), Public key Infrastructure(PKI), Digital Signatures and certificate (X.509), Applications of Cryptography. | 7 | CO2 |



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## Syllabus :

| Module No. | Detailed Content | Hrs. (39) | CO |
|---|---|---|---|
| 3 | Cyber security safeguards- overview, access control, audit, authentication, biometrics, denial of service, ethical hacking, firewalls, intrusion detection systems, security policy and threat management. | 6 | CO3 |
| 4 | Introduction to digital forensics, cybercrime scene investigation: Identifying and documenting the scene and evidence, maintaining the chain of custody, cloning of evidence, live and dead system forensic, perseveration of evidence, hashing concepts to maintain the integrity of evidence, examination and analysis of evidences, report generation. | 7 | CO4 |



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## Syllabus :

| Module No. | Detailed Content | Hrs. (39) | CO |
|---|---|---|---|
| 5 | Data recovery of tempered data, examining window registry and recycle bin, analysis of metadata, Restore points and shadow copies. Tools: Disk Imaging, Data Recovery, Vulnerability Assessment Tools, Encase and FTK tools, Anti Forensics and probable counters, retrieving information, data recovery tools. | 8 | CO5 |
| 6 | Cyber Security Regulations, Roles of International Law, the state and Private Sector in Cyberspace, Cyber Security Standards. The INDIAN Cyberspace, National Cyber Security Policy 2013. Basics of Indian Evidence ACT IPC and Criminal Procedure Code, Electronic Communication Privacy ACT, Legal Policies. | 5 | CO6 |



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## Textbooks :

1. Cryptography and Network Security, Atul Kahate, Tata Mc Graw Hill.

2. Behrouz A. Ferouzan, “Cryptography & Network Security”, Tata Mc Graw Hill.



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## Reference Books: :


1. Kevin Mandia, Chris Prosise, Matt Pepe, Incident Response and Computer Forensics, Tata McGraw - Hill, New Delhi, 2006.
2. Sunit Belapure Nina Godbole, Cyber Security, Wiley India Pvt. Harish Chander, Cyber Laws and It Protection, PHI Publication





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**Module 1:**

# Lecture 1: Overview of Cyber Security



## Cyberspace

Cyberspace is a time-dependent set of interconnected information systems and the humans that interact with these systems.

-Ottis and Lorents (2010)

• It is a dynamic, evolving, virtual, connected, multilevel ecosystem of physical infrastructure, software, regulations, processes, and interactions influenced by an expanding population of contributors … who represent the range of human intentions.

• Cyberspace is fundamentally dependent on technical advancement and innovation. All digital interactions in this space, including sending emails, visiting websites, and using social media are part of cyberspace.



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# Cybersecurity

- Cybersecurity is the practice of deploying people, policies, processes and technologies to protect organizations, their critical systems and sensitive information from digital attacks.

- A strong cybersecurity strategy can provide a good security posture against malicious attacks designed to access, alter, delete, destroy or extort an organization's or user's systems and sensitive data.



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# Importance of Cybersecurity

- To keep information, data, and devices private and safe.
- To detect and prevent cyber attacks.
- To prevent unauthorized access.
- Improves the recovery techniques of data even if attack cannot be prevented.



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## Information Security

- The International Standards Organization (ISO) (2014) defines it as: “Preservation of **confidentiality, integrity and availability** (CIA) of Information.

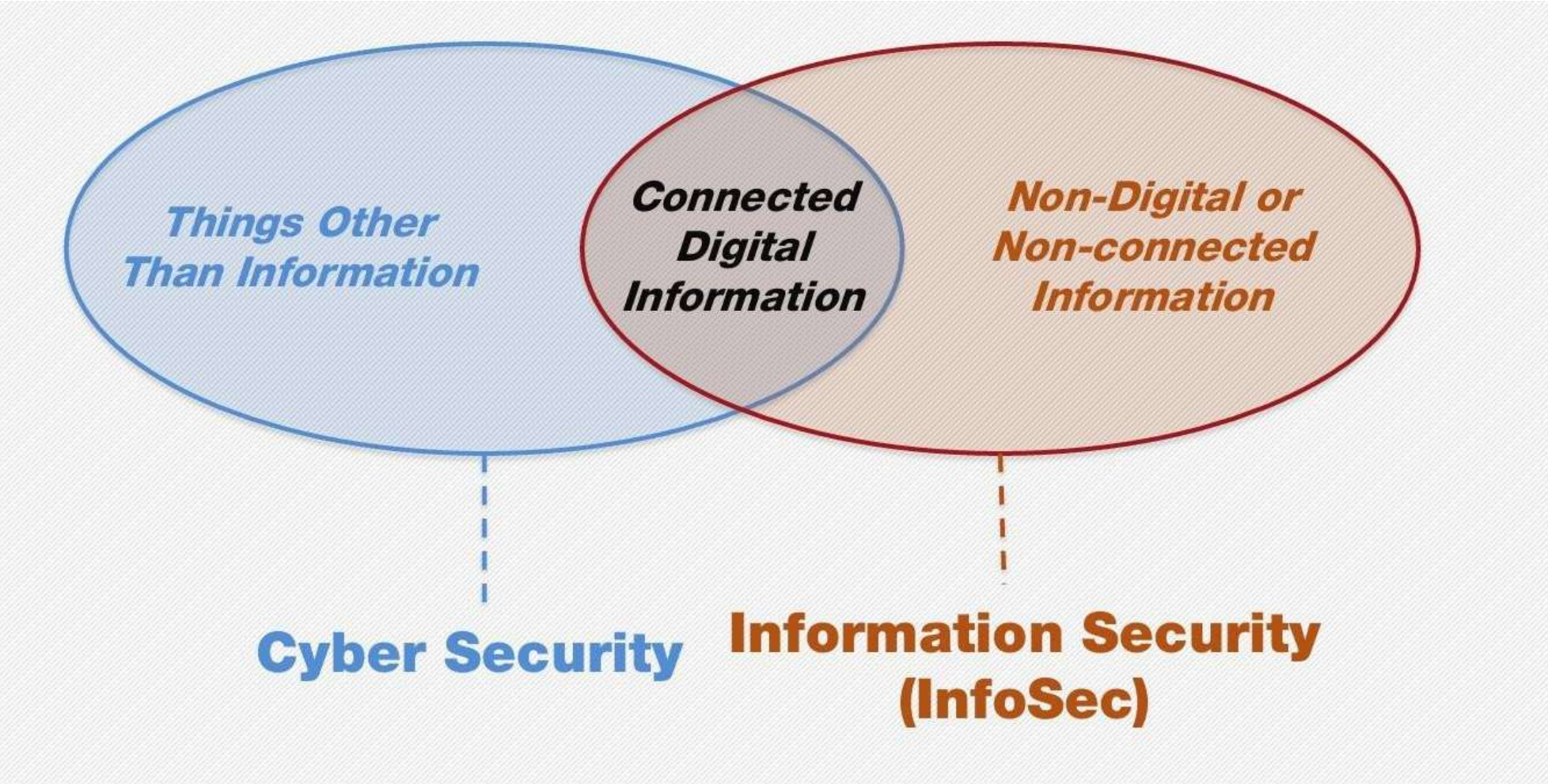
- In addition, other properties such as **authenticity, accountability, non-repudiation and reliability** can also be involved.”





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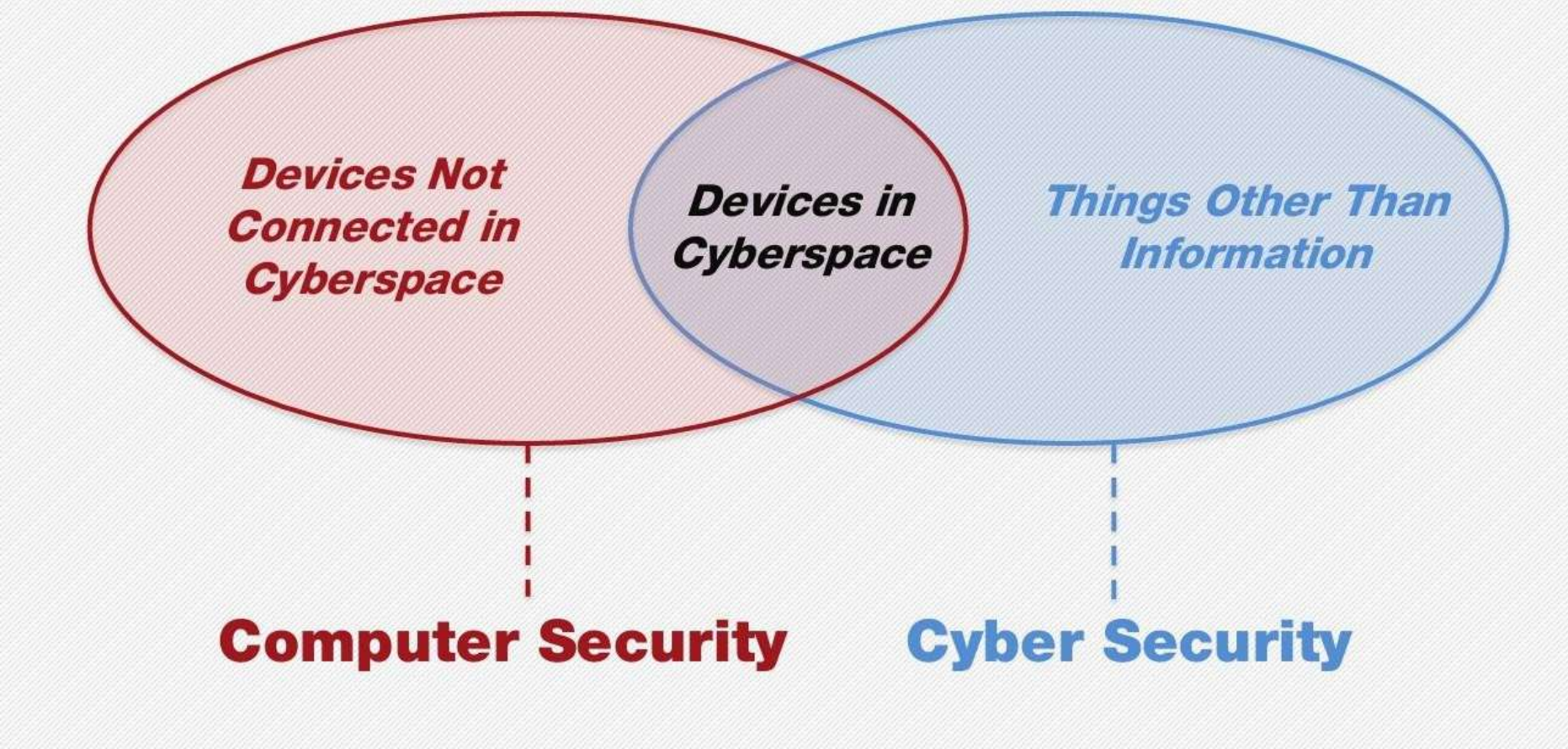
# Cyber Security Vs Information Security






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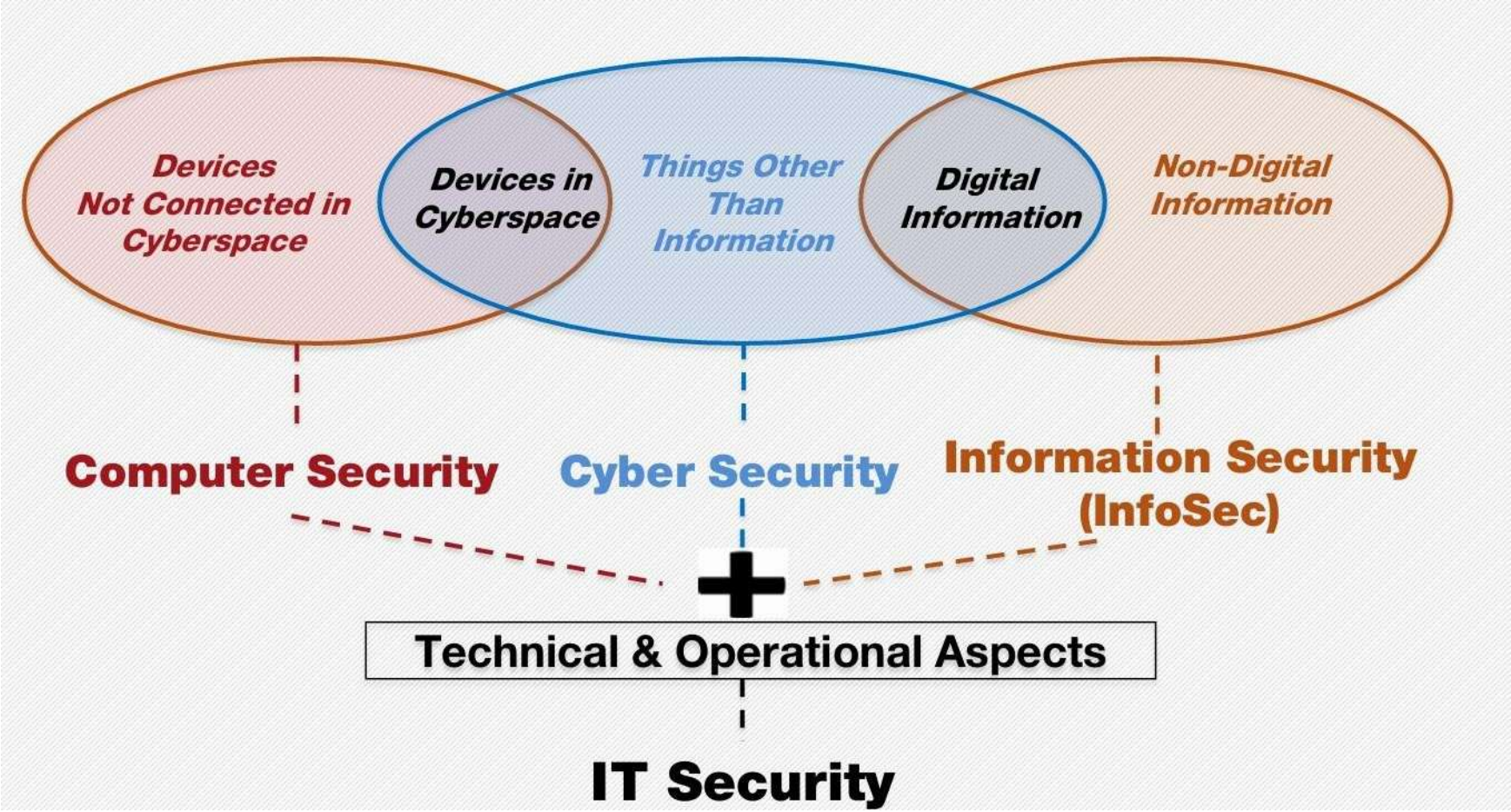
# Cybersecurity vs. Computer Security






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# IT Security






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# IT Security vs. InfoSec






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## Security Goals: The CIA Triad




Fig: CIA triad



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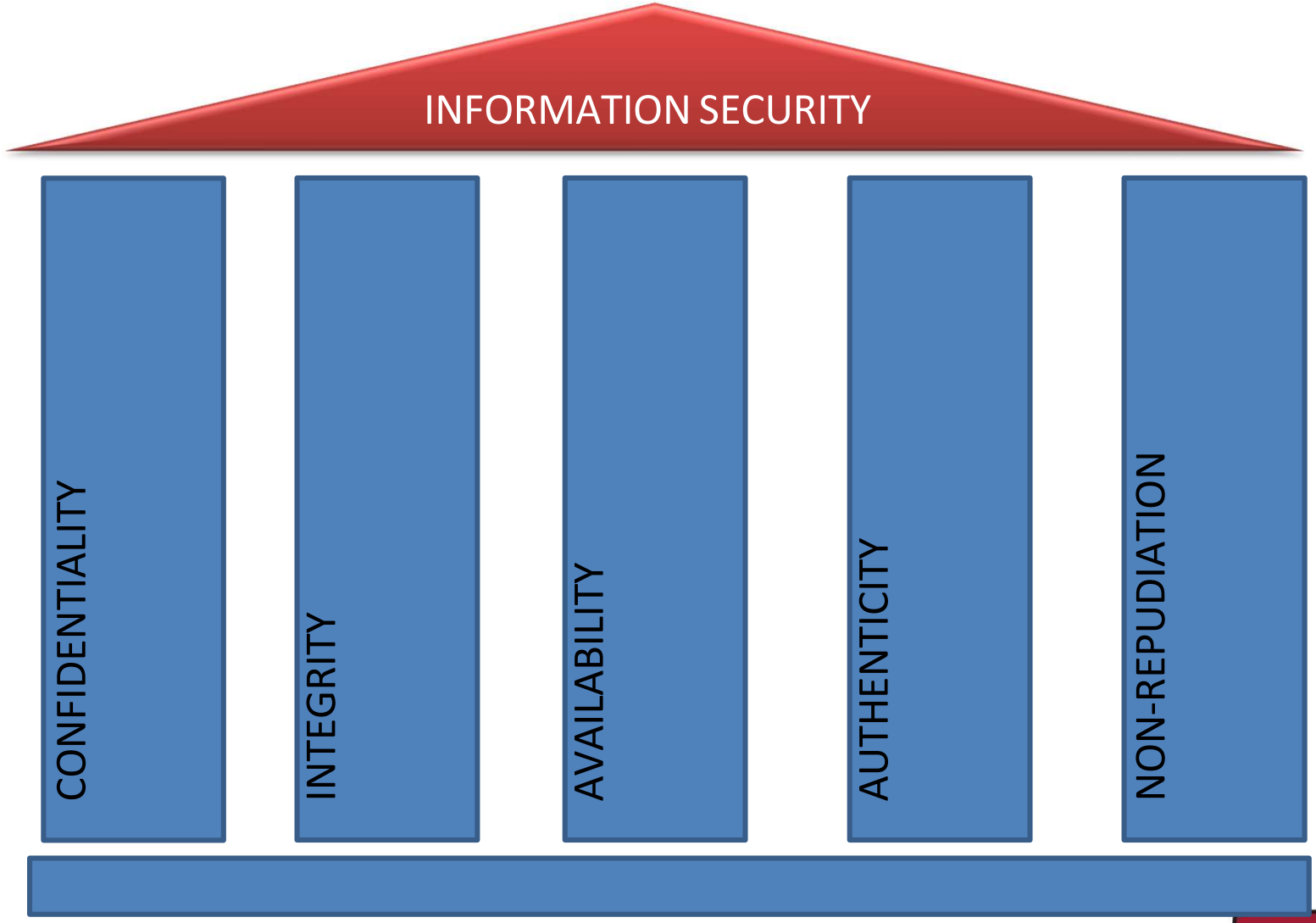
# The CIA Triad

- **Confidentiality**: Confidentiality ensures that information is accessible only to those who have the proper authorization and need to know. It prevents unauthorized access to sensitive data and aims to keep it confidential. Measures to maintain confidentiality include encryption, access controls, and user authentication.
- **Integrity**: Integrity guarantees that data remains accurate and trustworthy throughout its lifecycle. It protects against unauthorized alterations, tampering, or corruption of information. Data integrity is maintained through measures such as data validation, checksums, and digital signatures
- **Availability:** Availability ensures that data and systems are accessible and operational when needed. It safeguards against disruptions, downtime, or denial of service. Availability measures include redundancy, fault tolerance, backup systems, and disaster recovery planning.



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# Pillars of Information Security






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## Cybercrime

- It is a criminal activity in which computer, networks are a tool, a target or a place of criminal activity.
- It is also defined as an illegal activity involving a computer, a system and its activity.
- Cyber crime happens intentional not accidental.
- In cybercrime a computer may act as a target or as a tool.



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## Examples of Cybercrime

- Identity of theft due to fraudulent transactions.
- Wide spreading of computer malwares.
- Salami slicing is the practicing of stealing money repeatedly in small quantities.
- Making and digitally distributing child pornography.
- Unauthorised access to or modification of program or data.
- Software piracy.



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## Internet Governance

- Internet primarily is a Technological Phenomenon -- A set of rules and protocols, software and hardware.
- Internet governance is the development and application by Governments, the private sector and civil society, in their respective roles, of shared principles, norms, rules, decision-making procedures, and programmes that shape the evolution and use of the Internet.
- Actors are telcos, IETF, ISOC, RIRs, ICANN, Center for Internet and Society(India) etc.



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# Internet Governance

- **Telcos:**
- These are companies like Verizon, AT&T, or Vodafone that provide internet service to users. They are the infrastructure providers that allow users to connect to the internet.

- **IETF (Internet Engineering Task Force):**
- The IETF is a large, open international community of network designers, engineers, and researchers focused on the evolution of the internet architecture and the smooth operation of the internet. They develop and publish technical standards and best practices that define how the internet works.

- **ISOC (Internet Society):**
- The Internet Society is a global non-profit organization that works to promote the open development, evolution, and use of the internet. It plays a role in shaping internet policy and advocating for an open and accessible internet for all.

# Internet Governance

- **RIRs (Regional Internet Registries):**
- RIRs are organizations that manage and distribute IP addresses within specific geographic regions. They are responsible for allocating IP address blocks to Local Internet Registries (LIRs) and end users within their respective regions. Examples of RIRs include ARIN (North America), RIPE NCC (Europe), APNIC (Asia Pacific), LACNIC (Latin America and the Caribbean), and AfriNIC (Africa).

- **ICANN (Internet Corporation for Assigned Names and Numbers):**
- ICANN is a non-profit organization responsible for coordinating the global internet's system of unique identifiers, including IP addresses and domain names. It ensures the stable and secure operation of the internet's address space and domain name system. ICANN also works with the IETF on developing internet standards.

- Center for Internet and Society (India): This Indian non-profit organization conducts interdisciplinary research on internet and digital technologies. It focuses on policy and academic perspectives, addressing issues related to internet governance, privacy, and digital rights.

# Policy Topics in Internet Governance

- Cybersecurity
- Digital Trade
- Free Expression Online
- Privacy & Surveillance
- Internet of Things
- IG Institutions
- Internet Identifiers
- Geopolitics of IG



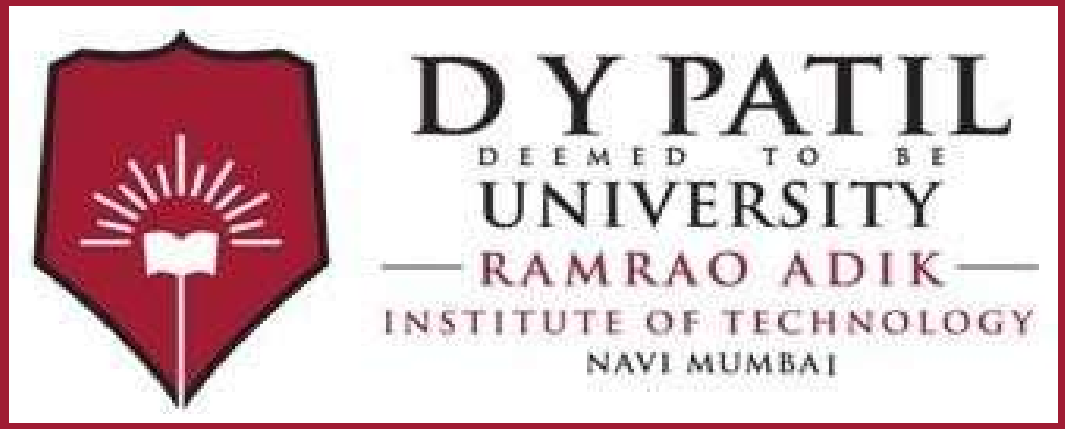
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## Challenges in Internet Governance

- The pace and changing nature of the internet.
- The internet as part of digitalisation.
- The concentration of digital power.
- Digital geopolitics (and the environment) .
- Shaping the digital future.
- The future of regulation.
- Multilateralism and multistakeholderism.
- Participation in decision-making.



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# Thank You